

“It just kind of shows that I went somewhere”: An Exploratory Study of Fitness Data Sharing

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Abstract—The sharing of curated fitness data posts occurs frequently on fitness-focused social platforms such as Strava and on general social media platforms such as Instagram, which is a novel context for visualization. To better understand the process of sharing and designing fitness data posts, as well as the role of visualization within them, we conduct a constructivist grounded theory study. We conduct and analyze 18 semi-structured interviews with fitness data sharers. From our analysis of the data, we find three novel characteristics of fitness data sharing: (i) the role of visualization as providing proof that an individual did an activity, (ii) the importance of expressing individuality in posts, and (iii) design conformity to cultural norms. We also derive a set of design implications, including a need for more options for visualizations for activities without routes, more user control in fitness data sharing platforms, and maintained ease of use while increasing customization options.

Index Terms—Visualization, fitness, social, communication.

I. INTRODUCTION

USING health data to support fitness progress is common [1] due to the ease of data collection through modern smart phones, watches, rings, and other devices. These devices record a variety of types of data, including the route an activity follows, the activity duration, the user’s heart rate, and the user’s calories burned, with different devices supporting different data for different activities. Applications to help wearers interpret their data, both from device developers themselves such as Garmin or third parties such as Bevel, are also readily available. These applications typically display the data in a variety of ways, including textually, numerically, and visually. They incorporate a variety of visualization idioms, such as geographic maps with routes shown on them, line charts, and bar charts. Beyond self-analysis, many modern fitness data applications also support the sharing of fitness data, either through in-app, fitness-focused social ecosystems or through exports of graphics to be shared elsewhere on Instagram and other general social media platforms. Figure 1 shows five examples of fitness data posts and exported graphics.

The sharing of fitness data represents a unique context for visualization research. It is *frequent*: users share very often, with some sharing multiple posts per day; it is *curated*: sharers have the flexibility to customize their posts; and it is *social*: the posts, including the visualizations, are shared

on online social platforms. While there are some similar contexts, none share all three of these qualities. For instance, episodic overviews, like the music listening platform Spotify’s “Wrapped” campaign and those inspired by it [2], are often *curated* and shared *socially*, but not frequently, as many of these overviews are only available once a year. Similarly, performance in daily games like Wordle and games inspired by it are *frequently* shared socially, but they are typically shared in a standard format without sharer curation. Likewise, the analysis of health data, such as sleep data [3] or menstrual health data [4], is often conducted *frequently* and using *curated* visualizations, although it is not shared socially.

While the fitness data sharing (FDS) context is niche, we argue that it may be a harbinger of similar visualization contexts which will become more common in the future. FDS became popular early due to the availability of devices for easily recording data relevant to activities such as GPS-enabled watches, in tandem with discursive social platforms for sharing this data [5]. Nowadays, improvements to devices and apps have led to them requiring significantly less technical proficiency, enabling a broader population to engage with them. We observe this and other similar trends occurring beyond FDS. In general, the availability and quantity of personal data is increasing, both through the number and diversity of features on wearable devices and the increase in passive collection of user interaction data on many platforms across multiple domains [2]. Moreover, the ability for people to access their data is also increasing with the advent of data protection laws which mandate access to users’ personal data [2].

We also note efforts to increase visualization ability in the general population, and in particular efforts to understand and improve skills such as the construction of visualizations [6], [7]. Software for making expressive visualizations continues to become more readily available and easy to use. Finally, sharing data visualizations socially is becoming increasingly popular, as is visible through the explosion of platforms developing annual shareable episodic reviews such as Spotify’s “Wrapped”, Steam’s “Year in Review” of video game play, YouTube’s “Recap” of video watching, the social media platform LinkedIn’s “Year in Review”, and the social fitness platform Strava’s “Year in Sport”, which may facilitate reflection and sensemaking [8].

To better understand the design and design process of FDS posts, and in particular the role of visualization, we conducted a semi-structured interview study. We followed the methodology of constructivist grounded theory [9] informed by others who have conducted grounded theory studies in HCI [10]. We conducted and analyzed 18 semi-structured

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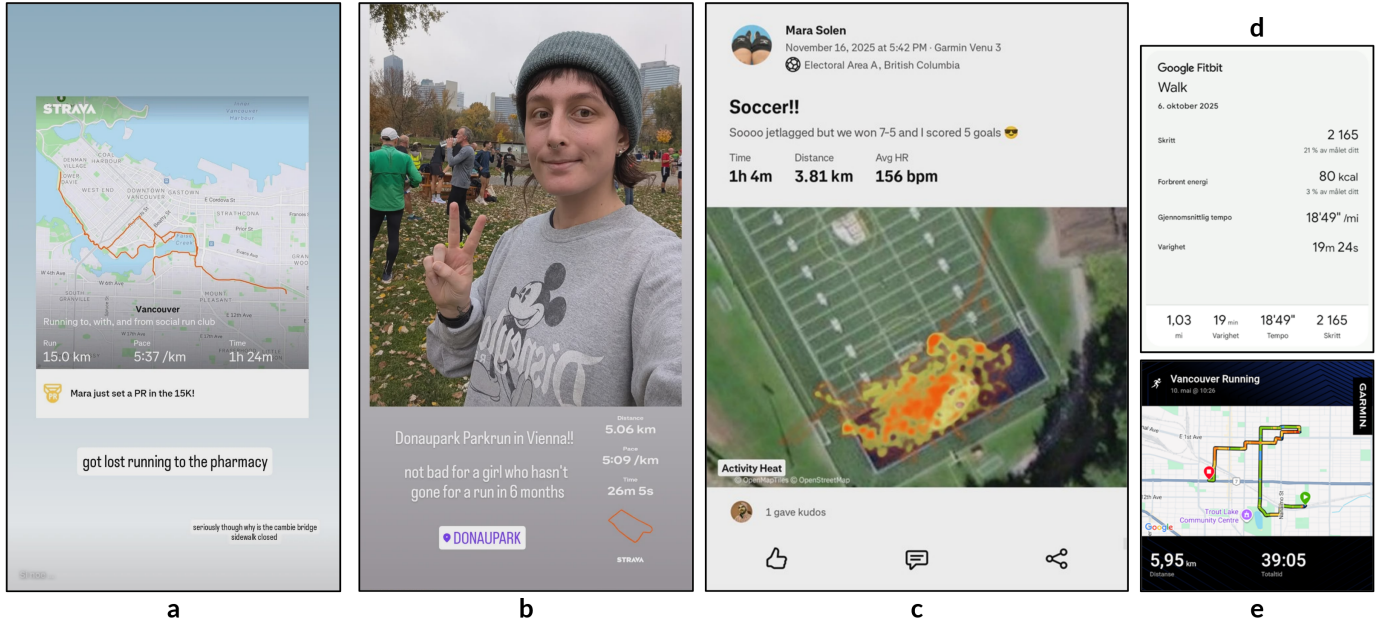


Fig. 1. Five examples of FDS posts and exports from the first author. a) Instagram Story with a full map export from Strava. b) Instagram Story with a transparent-background outline-only route export from Strava. c) Strava in-app post with geographic heatmap. d) FitBit export showing 7 unique metrics. e) Garmin export showing route with embedded pace data

interviews with fitness data sharers, to understand what people post within this context, why they post it, who they post it for, and what they wish was better about posting.

Our primary contributions are the results of our semi-structured interview study:

- First-stage analysis results, drawing from our codebook of 130 codes and 58 properties across 11 categories divided into 4 category groups.
- Second-stage analysis results:
 - Novel characteristics of fitness data sharing (FDS) which may apply to similar visualization contexts:
 - * The role of visualization as evidence-based *proof* for real-world activities, as opposed to communicating specific insights about the data.
 - * The importance of representing a sharer's activities, goals, and values.
 - * The pressure to conform to social norms of sharing platforms.
 - Design implications for FDS platforms:
 - * Greater variety of shareable visualizations and numerical metrics, especially for non-route activities, which are almost entirely unsupported.
 - * Increased user control over what is shared and its visibility to themselves and others.
 - * Maintaining ease of use while increasing options for customization.

Our secondary contribution is the suite of research artifacts we created, including a full codebook with both definitions and sample quotes for all codes, as well as anonymized full interview transcripts. These artifacts provide evidence for our analysis results and allow for alternative future analyses.

II. RELATED WORK

Existing literature contains related work on a variety of connected topics including personal analytics, sociological perspectives of social FDS platforms, studies investigating the roles of individual post components such as specific metrics and images, and tools for designers of and sharers using FDS applications. However, no existing work investigates the intersection of these topics, which is the design and design processes, and in particular the role of visualization, of fitness data posts on popular existing platforms.

A. Personal Informatics

There is a large body of work exploring personal analytics for fitness [11]–[13] as well as other topics including music listening [14], [15] and utility usage [16]–[18]. This work typically focuses on users' personal interactions with their data, but can still offer some insights related to the *personal data* aspects of personal data sharing.

Data agency, which refers to concerns about what information is collected and available to view, and effort, which refers to the amount of effort it takes to prepare visualizations, are two important concepts in personal informatics, and can influence whether tools are of interest to users [19]. Designing personal informatics tools that motivate users to sustain engagement is challenging, even when the tool is intended to directly improve the users well-being by supporting their health [20]. Disengagement with personal informatics is called lapsing, which can be both temporary or indefinite, and researchers have proposed many reasons for why it occurs [21]. Temporary reasons include forgetting to track their data, facing challenges maintaining their devices, intentionally choosing not to track specific instances, and intentionally suspending tracking behaviour due to a lack of interest or need. Some users

indefinitely lapse for a variety of reasons, such as achieving their goals, a lack of benefit to tracking, or a loss of curiosity in the data.

While this research on personal analytics overlaps with our findings, it often focuses on data management and integration of the information into the person's life, and does not cover the sharing of personal data in depth.

B. Sociology

There is a significant body of work investigating social fitness platforms such as Strava from a sociological lens. Much of this work focuses on motivations for using these platforms, with some finding positive effects of Strava on an individual's relationship with sports [22]–[24] and relationships with other people [25]. This effect is amplified further when the individual receives positive social feedback [23]. Existing work has also identified that Strava usage facilitates its own community [26], within which different social classes form based on fitness level [27], potentially leading to individuals feeling left out [26]. Within these communities, there is social pressure to post about completed activities [26], [28], as omitting a post can lead to the fear of missing out [26]. Felczak and Mirosław find that there are four frames of use for these platforms: social, hardcore, exploration, and training, and that different individuals have different motivations depending on their use [5], [29]. This social aspect of these fitness platforms may describe why fitness data sharers are motivated to sustain their engagement, in contrast to users of other personal informatics tools as mentioned in the previous section.

This body of work also describes lower-level details of social fitness platform use. Users compare their posts with others within their networks [5], [22], [30] to see what and how others are doing, however this comparison can lead to negative affect [26]. Some users also use these platforms to facilitate self analysis [30], [31], with that being their primary use. Self analysis can lead to a variety of affective outcomes [26], and in cases where the sharer is dissatisfied with their performance, they may remove the activity or make it private [22], [32]. Sharers on these platforms are aware that others can see their posts, with some finding this awareness motivating [5] while also being cognizant of how frequently they post [28].

While these sociological perspectives to social fitness platform use are directly relevant and informative to our work and our findings align with many of them, they are confined to general platform use, and do not cover sharers' design decisions.

C. Individual Post Components

Existing work has investigated the impacts of individual post components, both on general social media platforms as well as social fitness platforms. Images, which are a common component across many social platforms, are found to significantly increase a viewer's motivation to engage with a post [33], with photos getting the most likes and increasing the sense that the events described in the post really happened [34]. Selfies have received significant attention from the research

community, who have found that they are used to seek attention, communicate information, archive information, and entertain others [35].

Metrics, such as heart rate, step count, or pace, are another post component investigated in existing work. Metrics can be used to provide simple, high-level summaries or complex, low-level details. Although fitness devices capture the information as low-level details, social fitness platforms typically restrict the sharing of metrics to high-level summaries. Researchers have investigated how to share low-level details socially, however while doing so they identified a key challenge of maintaining user privacy over their data [36].

While social fitness posts can include both media and metrics, these studies do not consider the design of the entire fitness post or how these components are juxtaposed or overlaid with others, nor do they discuss visualizations.

D. Existing Tools

In 2020, Epstein et al. proposed Yarn, a tool for telling data-driven stories [37]. Their formative interview findings relate to our work, as users wanted to share a broad set of data types, such as images, text, and metrics, that emphasize aspects of the information that they found important. They also found that users preferred visual data formats and that sharers had a variety of goals with their posts, such as the goal of encouraging others. However, they found only modest success with the tool, and identified challenges, including that strict post templates constrained user creativity.

Wang et al. investigated the use of personal-data-driven digital stickers on ephemeral social media posts [38]. They discuss the differences between ephemeral and persistent posts, describing how ephemerality tends to be more private and playful, often documenting more mundane or everyday events. From their study, they find that the playfulness of the stickers made it easier and more enjoyable for users to share their information and start conversations with other users due to the perception of these stickers as less serious and more social than other platforms. They also found that sharing personal data increased accountability in the sharers, that sharers customized their posts with stickers for specific audiences, and that sharers felt it was important to align with the stylistic norms of the sharing platform. While this study is informative, it does not investigate an established community focused on a specific topic, as the stickers covered a variety of topics and were shared to more general audiences.

Epstein et al.'s 2015 design framework for social sharing in personal informatics [39] also overlaps with our work. They identify many reasons why people share on social media, and provide a design framework for designing tools for posting personal informatics based on an extensive review of prior literature.

Our work differs from these existing tools in that we are investigating the use of existing, successful tools with established communities to understand low-level design decisions of users, with a particular emphasis on the visualizations in the posts.

III. METHODOLOGY

In this study, we followed the methodology of constructivist grounded theory [9], conducting a series of semi-structured interviews about the sharing practices and experiences of 18 participants who share their fitness data online, informed by a review of fitness data posts on Instagram Stories and of existing platform features.

A. Positionality and Bracketing

The first two authors of this paper are embedded in fitness communities where FDS is common, including fitness class groups, sports teams, running groups, and online social fitness groups. Being embedded in these communities enabled the project through the identification of FDS as an interesting context for study through a visualization lens. It also enabled the sourcing of Instagram Stories about fitness data, and it provided access to and an increased trust level with interview participants. However, as both of these authors are regular sharers as well as visualization designers and researchers, they may think differently about the topic than those without visualization expertise, and as such have preconceptions and personal interests which are not reflected in the larger community.

We used a set of bracketing techniques to mitigate the deleterious effects of these preconceptions and personal interests [40]. The first author, who was the primary data analyst in all analyses, conducted initial reflexive journaling, where she wrote out an extensive description of her ideas about FDS. During the data collection and analysis, both the first and second authors wrote memos to document their developing understanding of the data. After each interview, the first two authors met to conduct peer debriefing, to further decant our developing understandings. Finally, all four authors met regularly to discuss the data collection and analysis, providing further opportunities to elicit and challenge preconceptions and biases.

B. Preparation

The first author reviewed public Instagram Stories with fitness data as well as features of FDS platforms to inform the design of the interview protocol and script.

In the fitness communities in which the first two authors are embedded, Instagram is a common place for the curating and sharing of fitness data information and graphics sourced from FDS platforms. The first author has online connections with local fitness leaders such as fitness class instructors and run club leaders who publicly share their own fitness data posts and re-share those of others in their fitness communities, with or without additional curation. The first author opportunistically took screenshots of Instagram Stories from these leaders from April 2024 to September 2025, leading to a collection of 54 screenshots of Stories created by 37 unique users. She then reviewed the screenshots to gain a preliminary understanding of what people include in fitness data posts.

The first author then explored the sharing features of four fitness data sharing platforms which were most commonly

used in the set of Instagram Stories: Strava, Garmin Connect, Apple Health, and FitBit. She collected screenshots of the shareable outputs and wrote descriptions of the features to further inform our interview design. See supplemental materials for a description of the platform features.

C. Data Collection

Based on author expertise and the reviews of the Instagram Stories and of the platform features, we developed our semi-structured interview script and protocol. The interview script had three sections:

- 1) participant demographics and orienting questions about fitness level, fitness activities, and general social media use;
- 2) questions about participant sharing practices and specific examples of shared posts;
- 3) and prompts about different features and points of dissatisfaction noted by previous participants.

We made small modifications to the interview script throughout the interview phase to improve flow and clarity for future participants. We asked for screensharing when we were unfamiliar with what the participant was describing.

We recruited interview participants through social media. The first author posted the flyer on Instagram and asked local fitness leaders to share it to their connections. We also advertised the flyer on the first and fourth authors' university department's Slack. Our focus on our local community allowed us to reach faster saturation, and is intended to describe a context grounded in a particular community rather than being globally representative.

Individuals who saw the flyer and were interested in participating in an interview filled out a pre-screening survey to verify that they met inclusion criteria. To be included in the study, respondents had to be nineteen years old or older, speak English, and have shared fitness data socially before. We also asked users to briefly characterize their fitness data sharing, such as what tools they used, how often they posted, and what fitness activities they partook in. We chose our initial participants to achieve diversity in these characterizations, and then later invited additional participants to further saturate areas where we found we lacked data based on memos and internal discussions. We excluded participants with established personal relationships with the interviewers. We conducted 2 pilot interviews with colleagues to refine our interview protocol and script which we exclude from the analysis. We then conducted 18 more interviews before reaching saturation. Participants included 9 women, 6 non-binary people, and 3 men ranging from 22 to 53 years of age who were all based in or recently moved from Vancouver, Canada. Their 28 different fitness activities included running (15), gym (10), walking (8), cycling (5), and 24 others. See supplemental materials for full participant demographics.

The first two authors participated in all interviews using pair interviewing, where one interviewer acts as the "driver" and focuses on asking the questions on the script, while the other interviewer acts as the "navigator" and focuses on taking notes and asking targeted follow-up questions [41]. We conducted

the interviews remotely over Zoom. The interviews lasted on average 60 minutes. We provided a \$25 CAD gift card to a fitness-related store for participating in the study. Zoom provided automatic transcriptions of the interviews which we manually edited where necessary. Our final interview dataset includes around 193,000 total words and over 1,000 minutes of interview audio. We supplement this dataset with memos taken by the interviewers both during each interview and in the debrief discussion immediately following each interview.

See supplemental materials for the full interview protocol and script as well as de-identified interview transcripts.

D. Data Analysis

We conducted a multi-stage coding process to analyze the Interviews. The first author conducted an initial review of all data, including memos and transcripts from the interviews. She then conducted in vivo coding [42] of the interview transcripts with the NVivo tool, highlighting snippets for further analysis. She created an initial codebook based on her initial review of the data.

She then coded the interview transcripts in depth, producing a set of codes grouped into categories, as well as properties of the categories. The properties are a middle level in between the category level and the code level, and serve to characterize the category. One code can be included in many properties and one property can involve many codes. The second author, who participated in all interviews, then joined the refinement iterations on the codebook, discussing with the first author and providing feedback which led to refinements [43]. Finally, the third and fourth authors joined the final refinement iterations of the codebook. As these two authors did not participate in the interviews, their feedback primarily led to clarity refinements.

To determine saturation, we recorded the number of new properties that emerged with each new interview. We chose properties over codes as the granularity of codes meant that several new codes could emerge without informing new properties. As we found only one new property from the last five interviews, we determined we had reached saturation. The final codebook contains 11 **categories**, spanning 58 **properties**, and 130 total **codes**. Each code is characterized by a definition, sample quote, and description of which participants the code applies to. Each property is characterized by a definition, list of associated codes, and which participant the property was first noticed in. The categories are further aggregated into 4 higher-level **category groups**: purposes, posts, perceptions, and desires. The entire codebook, from the first iteration to the fourth and final iteration, is included in supplemental materials.

We conducted a reflective second stage of analysis, drawing from the entire codebook, to produce a set of three novel characteristics of FDS and a set of three design implications for FDS platforms.

IV. RESULTS: FIRST-STAGE ANALYSIS

We now discuss the properties of the codebook, using individual codes and quotes only as examples as there are too many to exhaustively discuss. We discuss the eleven

categories according to four higher-level groups: purposes, posts, perceptions, and desires. We additionally note where our findings confirm existing work, meaning that our findings align with those of a previous study, or advance existing work, meaning that our work either provides additional depth to previously-covered topics or suggests new ideas which were previously uncovered. See Table I for the full set of properties.

A. Purposes

The Purposes category (Cat-A) contains the high level reasons for socially sharing fitness data. Participants mentioned a variety of reasons for sharing, confirming previous work [5], [23], [24], [26], [29]–[31], [44]. Property A1 describes how people share for themselves, and is associated with eight codes: sharing for self tracking (CA1), for accountability (CA4), to build new connections (CA6), to maintain connections (CA7), to remind themselves of their community and social connectedness (CA8), to keep people updated on their life and what they are doing (CA9), to allow people to check on their wellbeing (CA10), and to boast or show off (CA13). Beyond this point in this paper, we will not list all associated codes for each property; we only refer to a subset of associated codes, without labels, to describe a property.

People also share for the benefit of others who view that shared post (property A2), for example, to entertain them. Participants also mentioned sharing for a sense of community (A3), both fitness-specific and more general. They mentioned that sharing their fitness data helps them to both build and maintain community with those they have connected with online, and that viewers' interactions with posts reminded them of the support that their community provides. Finally, participants mentioned sharing to provide access (A4), to themselves or others. For example, sharing so that people can check on their mental and physical well-being without asking directly is a code pertaining to two properties, A2 and A4.

B. Posts

The Posts categories are the goals and desires that shape a person's posts (Cat-B), how much effort they put into posting (Cat-C), the frequency of their posts and how they choose what to post (Cat-D), the chosen format and included components in their posts (Cat-E), and the content that they cover in their posts (Cat-F).

Participants care about and seek to increase their post quality (B1), including their aesthetic quality, how engaging they are, and the inclusion of visuals. Participants were also cognizant of the quantity of posts they shared (B2), saying that they did not want to overwhelm others with too many posts, in particular if those posts were similar each time. These two properties relate to the next, which is that people care about how they contribute to others' feeds (B3). Participants mentioned a desire to post more high-level summaries rather than low-level details (B4). They also mentioned a desire to represent their personal values (B5). For example, multiple participants discussed the importance of transparency with fitness progress and how that impacted their posts.

Category	Label	Property	Definition
Purposes for Sharing (Cat-A)	A1	Shares for personal benefit	They share because it benefits them, the sharer.
	A2	Shares for others	They share because it benefits the people who see the shared post.
	A3	Shares for sense of community	They share to build, maintain, and be reminded of their community.
	A4	Shares for access	They share to have access to the data and provide access to others.
Posts Goals (Cat-B)	B1	Cares about quality of posts	They care about the perceived quality of their posts.
	B2	Cares about quantity of posts	They are hesitant to share too many posts, or too many similar posts.
	B3	Cares about feed contribution	They care about how the posts they make appear on and contribute to others' feeds.
	B4	Seeks summaries	They want high-level summaries, emphasizing they did the activity, not fine-grained details.
	B5	Wants to represent their values	They do not make as many posts as they'd like due to the effort required. They want their posts to represent their values.
Posts Effort (Cat-C)	C1	By platforms/types of posts	The platform and type of post (temporary, permanent) impact effort required to make post.
	C2	By activity	Certain activities take more effort to post about than others.
	C3	Comparing to others	They compare the amount of effort they put in to others they know.
	C4	Not posting due to effort	They do not make as many posts as they'd like due to the effort required.
Posts Frequency (Cat-D)	D1	General practice	They follow a typical pattern in how often and what activities they post.
	D2	Avoiding sharing	They avoid sharing in general or just specific activities.
	D3	By performance	Their performance impacts their choice of whether or not to post.
	D4	By activity	They limit their postings to certain types and/or intensities of activities.
	D5	By platform	Their posting patterns are different on different platforms.
	D6	Changes to frequency	Their posting patterns changed over time.
Posts Format (Cat-E)	E1	Sharing qualitative data	They share qualitative data like textual descriptions.
	E2	Sharing quantitative data	They share quantitative data like performance metrics.
	E3	Sharing visuals	They share visuals including visualizations and media.
	E4	About defaults	Participants describe whether or not they change default information in their posts.
	E5	Platform comparisons	They use different posting formats on different platforms.
	E6	By performance	Their post format is dependent on their performance.
	E7	Format changes	The format of their posts has changed over time.
Posts Content (Cat-F)	F1	Sharing context	They share details about their activity that aren't captured automatically.
	F2	Relating to performance	They share information about and reasons for their performance.
	F3	Sharing content for others	They include content that is meant to benefit others.
	F4	Sharing non-activity content	They share content that is not directly related to the activity.
	F5	Changes to content choices	Their content choices changed over time.
Perceptions of Viewers (Cat-G)	G1	Caring about sharing	Participants describe if they think people, them and others, care or not about fitness posts.
	G2	About specific people	Perceptions about how they think specific people will respond.
	G3	Fears	They have fears about posting.
	G4	Potential reactions	They consider potential reactions from others when posting.
	G5	Engagement	They consider how engaged viewers will be by their posts.
	G6	Changes to feelings	Their feelings about fitness data sharing have changed over time.
Perceptions of Metrics (Cat-H)	H1	Metric relevance	They find certain metrics more relevant to them than others.
	H2	Liking metrics	They like certain metrics.
	H3	Disliking metrics	They dislike certain metrics.
	H4	Uses of metrics	They use metrics to describe certain aspects of their activities.
	H5	Preferences for metrics	They have preferences for what metrics to share.
Perceptions of Visualizations (Cat-I)	I1	Describes activity	They find that visualizations help to describe what the activity was.
	I2	General enjoyment	They enjoy visualizations.
	I3	Information density	They find certain visualizations to have high or low information density.
	I4	Using maps to find specific info	They find maps are useful for sharing specific information like distance, routes, & location.
Perceptions of Platforms (Cat-J)	J1	Thoughts on Strava	Participants share thoughts on Strava in general.
	J2	Garmin is challenging to use	They find that Garmin is hard to use.
	J3	Compares Garmin and Strava	They make comparisons between Garmin and Strava.
	J4	Compares Strava and Instagram	They make comparisons between Strava and Instagram.
	J5	Aesthetics	They discuss the general and sharing-specific aesthetics of platforms.
	J6	Purpose	They discuss the purpose they associate with different platforms.
Desires for Features (Cat-K)	K1	Improved customizability	They want more customizability in existing features of fitness data sharing tools.
	K2	Flexibility with metrics	They want more flexibility with what metrics they can share.
	K3	Diversity to other sports	They want more diverse posting options for sports aside from running/cycling.
	K4	Diversity for sharing visuals	They want more diverse visualization options for sharing.
	K5	More features for Strava sharing	They want new sharing features in Strava.
	K6	Doesn't know how to improve	They don't know how to improve fitness data sharing platforms.

TABLE I
THE FULL TABLE OF PROPERTIES WITH DEFINITIONS, ORGANIZED BY CATEGORY.

“what will sometimes hold me back from using the Strava visual for that is that, like, moving time piece, which I feel isn’t representative. Like, obviously it would be great if it was, and it, you know, probably makes me look faster than I am, but I don’t want to, like, misrepresent that” (P09)

These points largely confirm previous work which describes high-level reasons for using FDS platforms [28], [45], [46].

Participants carefully considered the amount of effort that they were putting in to posting. They mentioned that different platforms and types of posts, such as ephemeral versus persistent, impact how much effort posting takes (C1). Some of these differences came from platform features, such as Strava’s automatic posting, while others came from the idea that longer-lasting, more broadly-viewed post types such as Instagram grid posts should be more polished than temporary, targeted post types such as Instagram Stories posted to a subset of followers. Participants also described how specific details of the activity impacted their effort levels (C2), with certain activities such as milestones, social activities, organized events, or activities with visuals such as media and visualizations warranting more effort than others. Finally participants explicitly compared how much effort they put in to posting to others they know (C3), and in some instances described how they make fewer posts than they would ultimately like due to the effort required (C4). C1 confirms SnapPI’s findings about the differences in how people view ephemeral and persistent posts [38], while C2-C4 advance existing work by providing further detail on sharer effort level considerations.

Some participants described their posting habits as general practices (D1), although these practices differed for different platforms (D5). Another general practice was the avoidance of sharing (D2), either entirely or just for specific activities. Users who described this practice mentioned making the decision of what to post based on both the activity types and intensities (D4) and their performance during the activity (D3). Finally, participants mentioned that their posting patterns had changed over time (D6), citing how established in their sport they were, how long they’d been using the platform, their follower count, and comments from others as influences. D1-D3 confirm existing work that describes general practices for sharing, including sharing avoidance in general, for specific activities, or when dissatisfied with performance [22], [32], while D4-D6 advance it by describing lower-level platform- and activity-based choices, and how these choices change over time.

Participants discussed a variety of components that they include in their posts, including both qualitative information (E1) and quantitative information (E2), often through visuals like images and visualizations (E3). They discussed whether they leave the default text and visible metrics or whether they change them (E4), for example by editing a post title. Participants’ post formats were influenced by both the platform (E5) and their performance with the activity (E6). Finally, participants mentioned changing their post formats over time, for example to mimic others:

“Also because I noticed that other people in this group also do that, so that’s definitely, like, I’m

trying to imitate that” (P12)

Existing work has discussed the inclusion of text, images, metrics, and visualizations in fitness posts at a high level, so E1-E3 confirm existing work [28], [33], [34], [36], however, we advance this knowledge with discussion of how people choose these components in properties E4-E7.

Participants mentioned a variety of types of content that they share in their posts. They share context that is not automatically captured by their devices (F1), including what the activity environment was like, who took part in the activity, and how they felt during the activity. They also shared additional information about their performance (F2), both to describe their performance and to justify it. Some participants mentioned including content that was not related to the activity (F4) or that was specifically included to benefit others (F3):

“It’s kind of a funny picture, so... I hope they would, you know, get a little laugh” (P03)

Finally, participants described how their content changed over time (F5), for example to adapt to their changing follower base, to imitate others, or to reduce publicly-available information. These findings all advance existing work, as no previous work has investigated the content of shared fitness data posts. While Moore et al. observed users including humour in self-annotations in a personal informatics system [20], they did not discuss the sharing of humorous annotations.

C. Perceptions

The Perceptions categories are how sharers think about viewers (Cat-G), metrics (Cat-H), visualizations (Cat-I), and platforms (Cat-J).

Considering viewers, participants discussed whether or not they cared about viewers seeing their posts, and whether or not viewers cared about their posts (G1). Some participants considered specific people and groups and how they might respond (G2), as some groups may be more interested or engaged. They also mentioned fears about what viewers might think of their posts (G3), with one participant stating:

“I know that there’s a part of me that’s insecure and, like, what if they see my pace, and they’re like, oh my god, this guy is molasses” (P18)

They also considered specific reactions that others may have when they post (G4) and whether others would engage with their posts (G5). Finally, participants mentioned that their feelings and perceptions of the viewers had shifted over time (G6). G1 and G3 confirm existing work that found that people differ in how much they care about others seeing their posts, with some being embarrassed to share [30], while the remaining findings advance it by providing depth on how sharers think about their viewers’ perceptions.

Participants had a variety of opinions about device-tracked metrics such as distance, heart rate, or pace. They discussed how some metrics can be more relevant than others (H1) based on the activity type and sharer goals, for example describing how the “power” metric, which is used for cycling, would appear sometimes for running activities. They also explicitly mentioned liking (H2) and disliking (H3) specific metrics,

with calories burned and heart rate appearing particularly contentious. Participants described the metrics as having different uses (H4), either being helpful for describing their physical ability or for helping to describe the activity they did. Finally, they discussed how they have preferences for how they want to share metrics (H5), such as preferring high-level summaries over low-level details. Sharers would choose to hide metrics for different reasons, including both to align with their personal preferences and to avoid seeing and sharing triggering information. While existing work has investigated feelings around topics such as calorie tracking [47] and user feelings around sharing low-level data [36] these findings otherwise advance existing work.

Participants enjoyed sharing visualizations (I2). They found visualizations helpful for describing their activity (I1), by showing that they truly completed it. One participant stated:

“It’s like a stamp of, like, I did this (...) And so, I do feel like it’s more real with that little map, or the little outline without the map.” (P05)

which shows how the visualization acts as proof of activity rather than a tool for communicating specific information about the data, and that sports without support for visualizations lack that proof. They also mentioned preferences for the information density of the visualizations (I3), stating that too little makes them uninformative while too much can be overwhelming. Finally, they described maps as particularly useful for communicating additional information (I4), such as a strong sense of how far they moved during their activity, the route they took which others may want to consider, and where they are in the world. Most of these findings confirm existing work showing that people enjoy visuals [33], [34], as well as considerations of information density [48], the use of visualization for storytelling [49], and communicating scale with visualization [50]. However, our finding that visualizations are sometimes most useful as a high-level proof that an individual truly did an activity advances previous communicative visualization work. This previously-unexplored use case for visualization is a novel finding.

With many platforms for curating and sharing fitness data being available, participants had considered the differences between them, primarily discussing Strava, Garmin, and Instagram. They compared Garmin and Strava (J3), finding that Garmin can be overwhelming and challenging to use (J2) but more information dense while preferring the aesthetic presentation of Strava (J5) and acknowledging its focus on specific sports (J1). They also compared Strava to Instagram (J4), mentioning how Instagram is more about storytelling and experience sharing while Strava is more technical (J6), with one participant saying:

“I think that, like, tying my [Instagram] to that would maybe make it feel... like, more performative and less enjoyable, whereas, like, tying Strava to it doesn’t feel that way” (P06)

Perceptions of Strava confirm existing work that found that Strava is viewed as serious [26], while the other findings advance existing work which has not discussed or compared these platforms in terms of sharing.

D. Desires

The Desires for Features category (Cat-K) covers a variety of improvements and additions to FDS platforms discussed by participants. They wanted increased flexibility, both in terms of stylistic customization (K1) and metric choice (K2). They also wanted more diversity for metrics and visualizations for non-route sports (K3) and for visualizations for all sports (K4). They asked for new features in Strava (K5), such as in-app posts with images and data shown in a single graphic, the ability to show data-driven comparisons with their overall fitness and past activities, and the ability to include music in their posts. Finally, our interview participants, very few of whom had expertise in design or visualization, often struggled to vocalize concrete ideas to improve their experience (K6), with one participant stating:

“I think I don’t even know... what I don’t know” (P18)

These findings all advance existing work; no previous study has investigated low-level design options for FDS platforms.

V. RESULTS: SECOND-STAGE ANALYSIS

We now discuss the results of our second-stage analysis: three novel characteristics of the FDS visualization context and three implications for design.

A. Novel Characteristics of the FDS Context

Many aspects of the FDS context align with general communicative visualization contexts. For example, the importance of engagement and understandability of the posts reflects goals in contexts such as informal education [51] and news [52]. However, we identified three novel characteristics of FDS: the role of visualization as proof, expressing individuality in shared posts, and conformity to etiquette and culture.

1) *Visualization as Proof of Activity*: For FDS posts with visualizations, we found that the common visualization tasks [53] may not be relevant, as multiple participants described their use of visualizations without mentioning any particular insight from the data. However, visualizations are still necessary in fitness data posts, as our interview participants found posts without visualizations to be less appealing, and they were less likely to post about activities without visuals. Beyond appeal, the visualizations provide more evidence that the activity was real than simply describing the activity in text or with numbers. For activities with routes, the visualizations generally show that the individual actually went outdoors for the activity rather than staying indoors and using a stationary machine like a treadmill or stationary bike. The data in the visualization, collected via a third party in the form of the fitness device, acts as a stamp of confirmation.

2) *Individuality*: Individuality is another key characteristic of FDS. Fitness data posts represent the individual sharing them, both in terms of what activities they engage in and their goals and values. Our participants differed in what activities they found relevant to post. For some, posting about gym workouts was uninteresting, as they used gym workouts to support their primary sport, while others who focused more on

gym workouts would post about them regularly. However, participants noted that they would be more interested in posting all types of activities if they had better visualization support. Beyond which activities a person posts, what information they include in their posts also aligns with who they are, including their personal values and what activities they partake in. We also found that a person's posts change as their values, interests, and activities change. When people begin practicing a new sport, they post about it more frequently and put more effort into the posts. Once that practice is more established, they promote the activity less in their posts. Our participants also mentioned how increasing the size of their follower base made them more self-conscious of their posts, and how that change led to an unfortunate reduction in the individuality represented in their posts.

3) *Conforming to Cultural Norms*: Finally, multiple of our participants mentioned an etiquette or culture for fitness data posts, and described how they conformed to it and expected others to conform to it. This culture influences both what to include in a post, and how often to post, with repetitive posts about activities such as cycling commutes being viewed negatively and perceived as "clogging up" the feed while posts about social activities like run club runs were seen as required. Our participants described being frustrated or annoyed when others did not follow these norms. We noted, however, that there are multiple *subcultures* within this culture. For example, one participant described how they posted less information over time because they saw others putting in less effort, and they were embarrassed about their effort level. In contrast, another participant described how they began training with a running group at a running track, and noticed how much detail the other runners included in their posts, and then increased the amount of information they included to conform. We note that not all participants chose to conform despite awareness of the social pressure, with one participant noting that their followers knew them as the type to post differently than others.

B. Implications for Design

From our study, we draw three implications for the design of FDS platforms.

1) *More Visualization and Metric Options*: FDS platforms should implement additional metric and visualization options for non-route sports. The high-level map visualization for activities with routes drives higher engagement with these activities, and these activities have a variety of additional options, including summative numerical descriptions like distance, abstract visual glyphs in the form of route outlines with no geographic map in the background, and concrete visualizations in the form of routes embedded into geographic maps with additional data such as pace embedded in the route line. Sports without routes typically have no visualization options at all, and the included metrics are generic to any activity, such as heart rate or calories burned. To improve the sharing experience of these activities, platforms should consider activity-specific metrics, abstract visual glyphs, and concrete visualizations to align with the functionality available for route-specific activities.

These platforms should also consider additional visualizations and metrics to support comparisons with overall fitness and previous activities. For example, enabling a user to share how their performance compares against recent activities or include metrics that are shown relative to other activities rather than in absolute units would support the types of information sharing that fitness data sharers are already doing. Fitness data platforms do support comparisons to overall fitness and previous activities as part of their personal analysis features, but these analyses are not yet easy to share.

Our participants also described the usefulness of concreteness. For those who are familiar with a sport, showing information in units such as meters or kilometers is most precise and therefore preferred. However, those who are unfamiliar may better understand if this information were represented in terms of ideas such as flights of stairs or football fields [50].

2) *Greater User Control for Metrics*: Our participants described feelings of frustration when discussing the metrics that FDS platforms include in posts. The default metrics are not always the ones that people want to show. However, the only way to control which metrics are shown, if it is possible at all to change them, is by hiding metrics and then checking what the system replaced them with. These platforms should allow sharers to explicitly choose which metrics they want to share. Sharers should also have control over the granularity of the shared metrics, as a person may be comfortable sharing an overall average but not the minute-by-minute details. Finally, while these platforms provide some control over how metrics are viewed by allowing a user to choose metric or imperial units, they could also enable features such as global hiding of metrics which a user might never want to see, in their posts or others. These increases to user control over metrics would help to alleviate privacy concerns and align posts with the activity, and user values [47] and goals.

3) *Increased Ease of Use*: FDS platforms have become increasingly easy to use, as noted by some of our participants, and it is important to maintain this trajectory. The pre-designed templates with a small set of options help to reduce the effort required to make posts, especially when the posts include visualizations as non-experts struggle to design their own. As platforms implement our other two design implications, they must maintain this ease of use. Some of our participants desired more options for stylistic customization through layout, colour, and font options, but they wanted a small set of options to choose from rather than full flexibility. Sharers may be happy to spend significant amounts of time customizing their posts, for example by changing styles or visible metrics, but the platform should allow users to create templates and profiles which they can design once and then persist for later re-use, as sharers are currently dissuaded by increasing their customization efforts due to frustration about needing to repeat their edits for every activity. None of our participants wanted full design toolkits embedded in the FDS platforms, possibly due to the existing availability of feature-full third-party tools.

VI. DISCUSSION

We now discuss the potential for our findings to apply to other similar contexts, methodological reflections, and limita-

tions and future work.

A. Extending to Similar Contexts

There are many areas with established online communities where users share information regularly, one example being video games [54]–[57]. Many video games have communities, with huge diversity between the purpose and type of interaction within these communities. Nintendo’s *Animal Crossing: New Horizons* enabled people to be social in creative ways, showcasing their individuality through digital creations [55]. Even in competitive fighting games and speedrunning, players can showcase individuality by finding success with unpopular strategies, with players becoming famous for doing so [58]. In massive multiplayer online games (MMOs) and team based games, players often hold specific roles within their individual communities and teams, often aligning with the player’s personal goals with the game [59]. Video game communities also contain cultural norms and pressure to conform, for example with players conforming to popular strategies, learning from others, and changing roles due to social pressure [54], [60].

These communities involve a significant amount of information and data sharing about designs, strategies, playthroughs, and even the games themselves, with social comparison being a motivator for sustained engagement. While this context currently involves frequent social sharing of information, many games already visualize information for the users [61], and there are online platforms for sharing data such as speedrunning times in consistent formats. Many of these communities would likely use visualization to communicate specific insights like how fast they were able to accomplish a task. However, in games where the primary challenge is completing a task at all, visualizations may serve as proof of completion, similar to our FDS interview participants. As more visualization tools arise, we may see video game players beginning to curate visualizations of their own game data on a large scale, similar to the FDS context.

We foresee video game communities, along with other social communities where information sharing is common, becoming more similar to the FDS context over time. If this prediction proves to be true, then they may share traits with the FDS context such as our novel characteristics.

B. Reflections on Methodology

We used constructivist grounded theory, although with some deviations. One notable deviation was the authors’ domain expertise and preconceptions and knowledge of existing work prior to beginning data collection. The original form of grounded theory guides users of the methodology to avoid the use of existing knowledge in the form of extant theory and personal domain expertise before data collection and analysis [62]. However, both Charmaz [9] and Furniss et al. [10] note that careful use of existing knowledge can support the research without introducing significant bias. As mentioned in Section III, it is our proximity to the domain which enabled this project, both in project identification and participant recruitment. We found bracketing to be helpful in reducing the issues with this proximity, and we suggest that

others in visualization consider bracketing techniques reflexive journaling and peer debriefing when working on domains in which they have expertise [40].

When conducting studies about design, participants without design expertise may struggle to actively think about what they do and why they do it. Furniss et al. suggest that researchers overcome this challenge by proposing ideas and discussing them with participants, rather than following standard grounded theory advice and keeping data collection fully open ended to avoid bias [10]. This finding aligns with visualization as well, as many people lack expertise in visualization design, and as such struggle to think of potential designs aside from those they are already familiar with. For our interview study, we attempt to combine both guidelines by first asking open-ended questions to gather unbiased participant ideas, get to know the participant, and gain rapport with them, before later providing specific prompts and discussing. We found that this combination enabled us to collect diverse opinions without needing to avoid discussion of ideas that the participant had not considered.

Finally, we used a modified form of pair interviewing [41]. We switched which interviewer held the driver and navigator roles between each interview as the two interviewers had subtly different domain expertise and providing both interviewers the opportunity to ask different types of questions enabled broader discussions. We also decided to let the driver keep track of the time during the interview, both because they were the one focused on the interview script and to avoid awkward interruptions from the navigator.

C. Limitations and Future Work

Our study is intentionally grounded in a particular community, which provides many benefits to the study such as increased rapport with participants and increased understanding of the data through analyst familiarity. We recruited most of our participants through Instagram, which we found was the most popular general social media platform for sharing fitness data in this community based on author familiarity. Additionally, none of our participants mentioned sharing fitness data posts to other general social media platforms. We also focused on Strava when considering in-app posts, which is the most popular fitness-focused sharing platform. However, it is possible that there are fitness data sharers who use other general and fitness-focused platforms, and that their use differs from that of our participant pool. Further, we did not aim for globally representative results, although the results of our qualitative exploration could enable broader quantitative studies in the future.

We made observations during our interviews that we determined were outside of the scope of this study, however they may make for interesting future work. We noted that multiple of our participants recorded their fitness data on multiple devices simultaneously, and we were unable to find out why. We also confirmed findings of existing work showing that some people share posts publicly on platforms like Strava but claim to use the platform exclusively for self tracking, which we did not explore further. We focused our analysis on the sharer

perspective, but it may be interesting to investigate viewer perceptions, for example by showing people others' posts and discussing with them. Finally, we note that our participants did not align in many different ways, and future work could attempt to distill specific traits of fitness data sharers, then conduct a quantitative study to look for correlations between these traits and construct sharer types.

VII. CONCLUSION

We conducted and analyzed 18 semi-structured interviews to produce a codebook with 130 codes and 58 properties that provides insights into the novel visualization context of FDS. Some of our analysis findings confirm existing work, and others advance it by providing additional depth to previously-covered topics or suggesting new ideas. We also contribute a set of novel characteristics that pertain to the FDS context and a set of design implications for FDS platforms. We find that visualization has a novel role in this context, as a visual stamp of proof that a sharer actually did an activity, and make suggestions on how to support this role for posts about non-route activities, where visualizations are currently under-supported. Our supplemental materials include our full codebook containing the codes with definitions and sample quotes and properties with definitions. They also include a document with additional post and export figures, platform feature collection details and descriptions, saturation details, an explanation of the relationships between categories, properties, and codes, and a description of each contributor's role in this work using the CRediT system [63]. They also include our interview protocol and script, participant demographic details, and all 18 de-identified interview transcripts. Supplemental materials are available at <https://osf.io/rma7w/>.

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REFERENCES

- [1] Y. Bhargava and J. Nabi, "The opportunities, challenges and obligations of Fitness Data Analytics," *Procedia Computer Science*, vol. 167, pp. 1354–1362, 2020.
- [2] T. J. Davidson, E. Lee, and E. Wall, "Spotify warped: Reshaping personal informatics via music listening, casual users, passive data and episodic reflection," *ACM Conf. Human Factors in Computing Systems (CHI)*, 2026.
- [3] Z. Liang, B. Ploderer, W. Liu, Y. Nagata, J. Bailey, L. Kulik, and Y. Li, "SleepExplorer: a visualization tool to make sense of correlations between personal sleep data and contextual factors," *Personal and Ubiquitous Computing*, vol. 20, no. 6, pp. 985–1000, 2016.
- [4] G. Lin, P.-W. Lessard, M. N. Le, B. Li, F. Chevalier, K. N. Truong, and A. Mariakakis, "Functional design requirements to facilitate menstrual health data exploration," in *ACM Conf. Human Factors in Computing Systems (CHI)*, 2024, pp. 1–15.
- [5] D. J. Rivers, "Strava as a discursive field of practice: Technological affordances and mediated cycling motivations," *Discourse, Context & Media*, vol. 34, p. 100345, 2020.
- [6] L. W. Ge, Y. Cui, and M. Kay, "AVEC: An assessment of visual encoding ability in visualization construction," in *ACM Conf. Human Factors in Computing Systems (CHI)*, 2025, pp. 1–16.
- [7] M. Solen, "Scoping the future of visualization literacy: A review," in *IEEE VIS Wkshp. Visualization for Communication (VisComm)*, 2022.
- [8] W. Li, J. Zhang, J. Ma, and P. Zhang, "From platform data to personal insight: How users make sense of and reflect on personalized social media annual recaps," *ACM Conf. Human Factors in Computing Systems (CHI)*, 2026.
- [9] K. Charmaz, *Constructing grounded theory: A practical guide through qualitative analysis*. Sage, 2006.
- [10] D. Furniss, A. Blandford, and P. Curzon, "Confessions from a grounded theory PhD: experiences and lessons learnt," in *ACM Conf. Human Factors in Computing Systems (CHI)*, 2011, pp. 113–122.
- [11] C. Fan, J. Forlizzi, and A. K. Dey, "A spark of activity: exploring informative art as visualization for physical activity," in *ACM Conf. Pervasive & Ubiquitous Computing (UbiComp)*, 2012, pp. 81–84.
- [12] S. Consolvo, D. W. McDonald, T. Toscos, M. Y. Chen, J. Froehlich, B. Harrison, P. Klasnja, A. LaMarca, L. LeGrand, R. Libby et al., "Activity sensing in the wild: A field trial of UbiFit garden," in *ACM Conf. Human Factors in Computing Systems (CHI)*, 2008, pp. 1797–1806.
- [13] I. Li, A. K. Dey, and J. Forlizzi, "Using context to reveal factors that affect physical activity," *ACM Trans. on Computer-Human Interaction (TOCHI)*, vol. 19, no. 1, pp. 1–21, 2012.
- [14] D. Baur, F. Seiffert, M. Sedlmair, and S. Boring, "The streams of our lives: Visualizing listening histories in context," *IEEE Trans. Visualization & Computer Graphics (TVCG)*, vol. 16, no. 6, pp. 1119–1128, 2010.
- [15] R. Dias, M. J. Fonseca, and D. Gonçalves, "Interactive exploration of music listening histories," in *Conf. Advanced Visual Interfaces (AVI)*, 2012, pp. 415–422.
- [16] L. Bartram, J. Rodgers, and R. Woodbury, "Smart homes or smart occupants? Supporting aware living in the home," in *IFIP Conf. Human-Computer Interaction*. Springer, 2011, pp. 52–64.
- [17] M. Chetty, D. Haslem, A. Baird, U. Ofoha, B. Sumner, and R. Grinter, "Why is my Internet slow? Making network speeds visible," in *ACM Conf. Human Factors in Computing Systems (CHI)*, 2011, pp. 1889–1898.
- [18] E. Costanza, S. D. Ramchurn, and N. R. Jennings, "Understanding domestic energy consumption through interactive visualisation: a field study," in *ACM Conf. Pervasive & Ubiquitous Computing (UbiComp)*, 2012, pp. 216–225.
- [19] D. Huang, M. Tory, B. A. Aseniero, L. Bartram, S. Bateman, S. Carpendale, A. Tang, and R. Woodbury, "Personal visualization and personal visual analytics," *IEEE Trans. Visualization & Computer Graphics (TVCG)*, vol. 21, no. 3, pp. 420–433, 2014.
- [20] J. Moore, P. Goffin, J. Wiese, and M. Meyer, "Exploring the personal informatics analysis gap: 'There's a lot of bacon'," *IEEE Trans. Visualization & Computer Graphics (TVCG)*, vol. 28, no. 1, pp. 96–106, 2021.
- [21] D. A. Epstein, A. Ping, J. Fogarty, and S. A. Munson, "A lived informatics model of personal informatics," in *ACM Conf. Pervasive & Ubiquitous Computing (UbiComp)*, 2015, pp. 731–742.
- [22] A.-M. Evgenieva, "Strava – social media or health app? empirical study report." Ph.D. dissertation, Strasbourg University, 2024.
- [23] R. Franken, H. Bekhuis, and J. Tolsma, "Kudos make you run! How runners influence each other on the online social network Strava," *Social Networks*, vol. 72, pp. 151–164, 2023.
- [24] H. C. Russell, C. Potts, and E. Nelson, "'If it's not on Strava it didn't happen': Perceived psychosocial implications of Strava use in collegiate club runners," *Recreational Sports Journal*, vol. 47, no. 1, pp. 15–25, 2023.
- [25] X. Gui, Y. Chen, C. Caldeira, D. Xiao, and Y. Chen, "When fitness meets social networks: Investigating fitness tracking and social practices on WeRun," in *ACM Conf. Human Factors in Computing Systems (CHI)*, 2017, pp. 1647–1659.
- [26] W. R. Smith, "Mobile interactive fitness technologies and the recreational experience of bicycling: A phenomenological exploration of the Strava community," Master's thesis, Clemson University, 2014.
- [27] E. Westlate, "'if you see me collapse, pause my Strava': Biopower, fitness data, and the anxious online performance of the fit body," *Sporting Performances: Politics in Play*, pp. 130–147, 2020.

- [28] P. W. Wozniak, A. Fedosov, E. Mencarini, and K. Knaving, "Soil, rock, and snow: On designing for information sharing in outdoor sports," in *Conf. Designing Interactive Systems (DIS)*, 2017, pp. 611–623.
- [29] M. Felczak and M. Filiciak, "Mobile health cycling: How Eastern European amateur cycling enthusiasts frame their experiences with Zwift and Strava," *Intl. Review for the Sociology of Sport*, p. 10126902251333566, 2025.
- [30] A. M. Williams, "King of the Mountain: A rapid ethnography of Strava cycling," Master's thesis, University College London, 2012.
- [31] J. Couture, "Reflections from the 'strava-sphere': Kudos, community, and (self-) surveillance on a social network for athletes," *Qualitative Research in Sport, Exercise and Health*, vol. 13, no. 1, pp. 184–200, 2021.
- [32] A. Alqhatani and H. R. Lipford, "There is nothing that I need to keep secret": Sharing practices and concerns of wearable fitness data," in *Symp. Usable Privacy and Security (SOUPS)*, 2019, pp. 421–434.
- [33] C. Johnston and W. E. Davis, "Motivating exercise through social media: Is a picture always worth a thousand words?" *Psychology of Sport and Exercise*, vol. 41, pp. 119–126, 2019.
- [34] E. Lowe-Calverley and R. Grieve, "Thumbs up: A thematic analysis of image-based posting and liking behaviour on social media," *Telematics and Informatics*, vol. 35, no. 7, pp. 1900–1913, 2018.
- [35] Y. Sung, J.-A. Lee, E. Kim, and S. M. Choi, "Why we post selfies: Understanding motivations for posting pictures of oneself," *Personality and Individual Differences*, vol. 97, pp. 260–265, 2016.
- [36] D. A. Epstein, A. Borning, and J. Fogarty, "Fine-grained sharing of sensed physical activity: A value sensitive approach," in *ACM Conf. Pervasive & Ubiquitous Computing (UbiComp)*, 2013, pp. 489–498.
- [37] D. A. Epstein, M. Dontcheva, J. Fogarty, and S. A. Munson, "Yarn: Adding meaning to shared personal data through structured storytelling," in *Graphics Interface 2020*, 2020.
- [38] D. Wang, M. Chheang, S. Ji, R. Mohta, and D. A. Epstein, "SnapPI: Understanding everyday use of personal informatics data stickers on ephemeral social media," *ACM Human-Computer Interaction (PACM-CHI)*, vol. 6, no. CSCW2, pp. 1–27, 2022.
- [39] D. A. Epstein, B. H. Jacobson, E. Bales, D. W. McDonald, and S. A. Munson, "From 'nobody cares' to 'way to go!': A design framework for social sharing in personal informatics," in *ACM Conf. Computer Supported Cooperative Work & Social Computing (CSCW)*, 2015, pp. 1622–1636.
- [40] L. Tufford and P. Newman, "Bracketing in qualitative research," *Qualitative social work*, vol. 11, no. 1, pp. 80–96, 2012.
- [41] D. Akbaba and M. Meyer, "Two heads are better than one": Pair-interviews for visualization," in *IEEE Conf. Visualization and Visual Analytics (VIS)*, IEEE, 2023, pp. 206–210.
- [42] J. Manning, "In vivo coding," in *The international encyclopedia of communication research methods*, 2017, p. 18.
- [43] N. Elmqvist and J. S. Yi, "Patterns for visualization evaluation," in *IEEE VIS Wkshp. Evaluation and Beyond - Methodological Approaches to Visualization (BELIV)*, 2012, pp. 1–8.
- [44] Y. Zhu, S. L. Dailey, D. Kreitzberg, and J. Bernhardt, "Social network workout": Connecting social features of wearable fitness trackers with physical exercise," *Journal of Health Communication*, vol. 22, no. 12, pp. 974–980, 2017.
- [45] G. Seidman, "Self-presentation and belonging on Facebook: How personality influences social media use and motivations," *Personality and individual differences*, vol. 54, no. 3, pp. 402–407, 2013.
- [46] D. Wang, J. Eng, M. Turpitka, and D. A. Epstein, "Exploring activity-sharing response differences between broad-purpose and dedicated on-line social platforms," *ACM Human-Computer Interaction (PACM-CHI)*, vol. 8, no. CSCW2, pp. 1–37, 2024.
- [47] C. C. Simpson and S. E. Mazzeo, "Calorie counting and fitness tracking technology: Associations with eating disorder symptomatology," *Eating behaviors*, vol. 26, pp. 89–92, 2017.
- [48] T. Munzner, *Visualization Analysis and Design*. CRC Press, 2014.
- [49] C. Tong, R. Roberts, R. Borgo, S. Walton, R. S. Laramée, K. Wegba, A. Lu, Y. Wang, H. Qu, Q. Luo *et al.*, "Storytelling and visualization: An extended survey," *Information*, vol. 9, no. 3, p. 65, 2018.
- [50] F. Chevalier, R. Vuillemin, and G. Gali, "Using concrete scales: A practical framework for effective visual depiction of complex measures," *IEEE Trans. Visualization & Computer Graphics (TVCG)*, vol. 19, no. 12, pp. 2426–2435, 2013.
- [51] M. Solen, N. Sultana, L. Lukes, and T. Munzner, "DeLVE into Earth's past: A visualization-based exhibit deployed across multiple museum contexts," *IEEE Trans. Visualization & Computer Graphics (TVCG)*, vol. 31, no. 1, pp. 952–961, 2024.
- [52] P. Sánchez-Holgado, C. Arcila-Calderón, and M. Frías-Vázquez, "The effect of interest and attitude on public comprehension of news with data visualization," *Frontiers in Communication*, vol. 8, p. 1064184, 2023.
- [53] M. Brehmer and T. Munzner, "A multi-level typology of abstract visualization tasks," *IEEE Trans. Visualization & Computer Graphics (TVCG)*, vol. 19, no. 12, pp. 2376–2385, 2013.
- [54] L. Rossi, "MMORPG guilds as online communities-power, space and time: From fun to engagement in virtual worlds," *Space and Time: From Fun to Engagement in Virtual Worlds (August 28, 2008)*, 2008.
- [55] L. Zhu, "The psychology behind video games during COVID-19 pandemic: A case study of Animal Crossing: New Horizons," *Human Behavior and Emerging Technologies*, vol. 3, no. 1, pp. 157–159, 2021.
- [56] R. Scully-Blaker, "Re-curating the accident: Speedrunning as community and practice," Ph.D. dissertation, Concordia University, 2016.
- [57] N. J. Winger, "Super smash bros. melee: Performance as community survival," Master's thesis, State University of New York at Buffalo, 2021.
- [58] "Smasher: Hungrybox," <https://www.ssbwiki.com/Smasher:Hungrybox>, accessed: 2026-04-02.
- [59] J. J. Mondal, A. Zahin, M. A. Manab, and M. Z. Hasan, "Does a support role player really create difference towards triumph? Analyzing individual performances of specific role players to predict victory in League of Legends," in *Intl. Conf. Computer & Information Technology (ICCIT)*, IEEE, 2022, pp. 768–773.
- [60] P. Vorderer, T. Hartmann, and C. Klimmt, "Explaining the enjoyment of playing video games: the role of competition," in *Intl. Conf. Entertainment Computing*, 2003, pp. 1–9.
- [61] V. Zammito, "Visualization techniques in video games," in *Electronic Visualisation and the Arts (EVA)*. BCS Learning & Development, 2008.
- [62] B. G. Glaser and A. L. Strauss, *Discovery of Grounded Theory: Strategies for Qualitative Research*. Routledge, 1999.
- [63] A. Brand, L. Allen, M. Altman, M. Hlava, and J. Scott, "Beyond authorship: Attribution, contribution, collaboration, and credit," *Learned Publishing*, vol. 28, no. 2, 2015.



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